# EX NO: 1 Estimation of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line from Primary Constants

**AIM:** Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line with

Resistance =10 Ω/Km,

Capacitance = 8 μF/Km,

Inductance=3.5 mH/Km,

Conductance =4 μmho/Km,

Frequency= 2 Khz

**SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

**PROCEDURE:**

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

**PROGRAM**:

```
clear;
clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
```

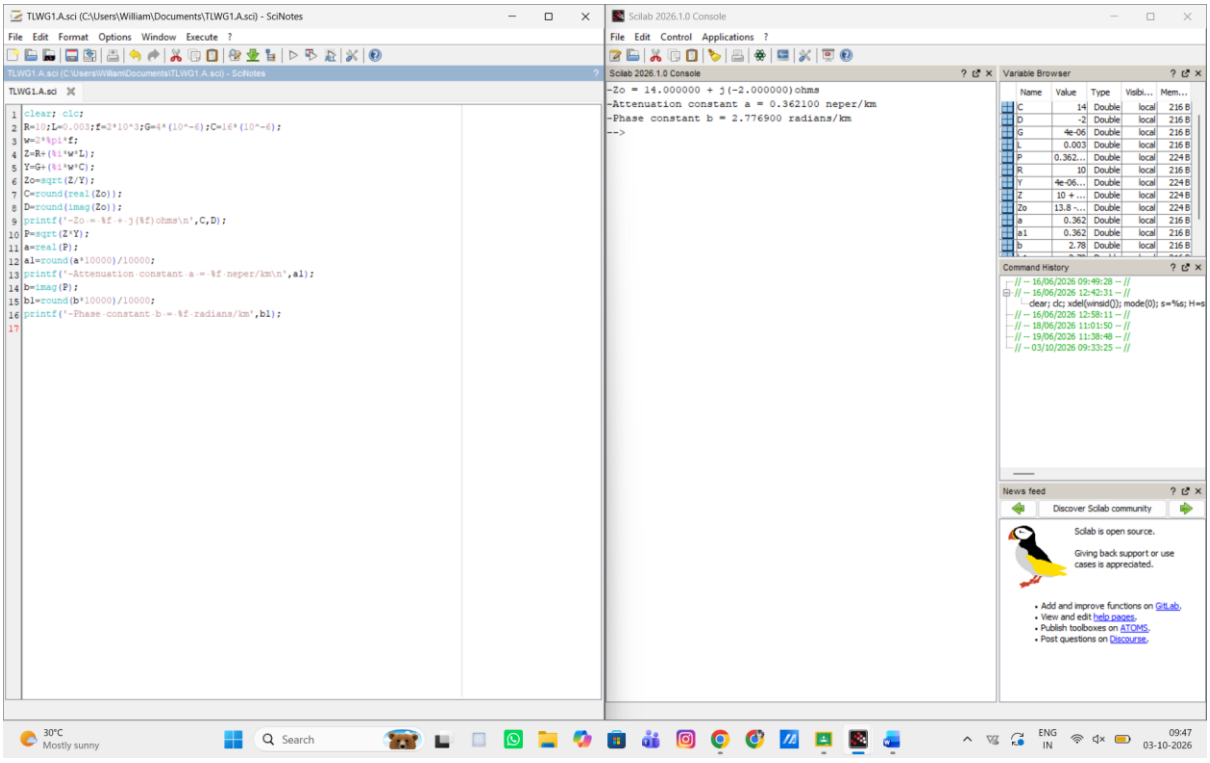
```
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

**OUTPUT:**

Constants L=3mH;

f=2000; G=.4*(10^-6);

C=8*(10^-6).



## Tabulation:

| Test Case | Simulation Results | | | Theoretical Results | |
|---|---|---|---|---|---|
| | $Z_o$ (Ω) | α (neper/km) | β (radians/km) | α (neper/km) | β (radians/km) |
| 1 | 14 − j2 Ω | 0.3621 | 2.7769 | 0.3621 | 2.7769 |

**RESULT:**

The characteristic impedance and propagation constant of a transmission line is calculated using SCILAB and the impact of primary constants on secondary constants are investigated.

**Test Case 2:**

**Examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.**

**AIM:**

To examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at afrequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L G and C values constant and calculate the Characteristic Impedance and Propagation Constant.Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

**SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

**PROCEDURE:**

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

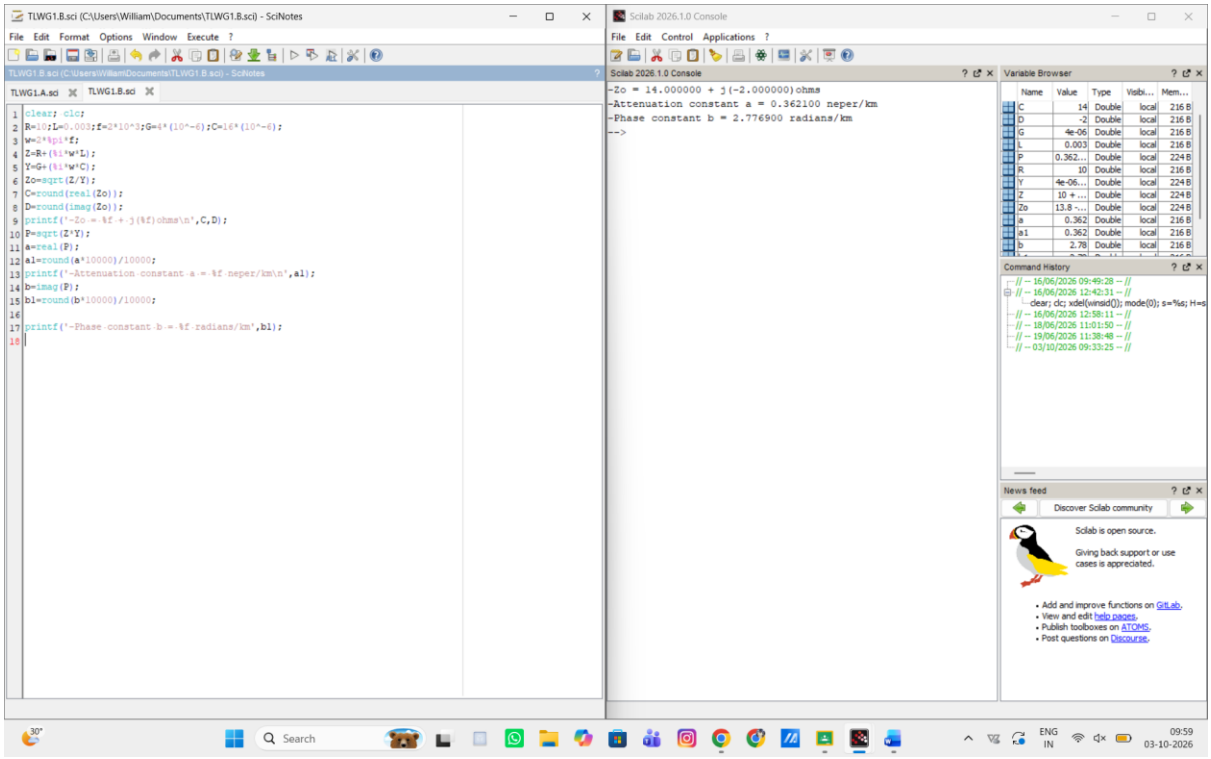
● Execute the program

● End the program

**PROGRAM:**

```
clear;
clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
```

printf('-Phase constant b = %f radians/km',b1);




## TEST CASE 2 — VARIABLE R

**Constants:**
L = 3 mH, f = 2.5 kHz, G = 0.4 × $10^{-6}$ S/km, C = 8 × $10^{-6}$ F/km

| S.No. | R (Ω) | $Z_0$ (Ω) | Simulation α (neper/km) | Simulation β (rad/km) | Theoretical α (neper/km) | Theoretical β (rad/km) |
|---|---|---|---|---|---|---|
| 1 | 10 | 19.47 − j2.04 | 0.2568 | 2.4470 | 0.2568 | 2.4470 |
| 2 | 20 | 19.78 − j4.02 | 0.5056 | 2.4854 | 0.5056 | 2.4854 |
| 3 | 30 | 20.24 − j5.90 | 0.7410 | 2.5438 | 0.7410 | 2.5438 |
| 4 | 40 | 20.82 − j7.64 | 0.9607 | 2.6162 | 0.9607 | 2.6162 |
| 5 | 50 | 21.47 − j9.27 | 1.1645 | 2.6978 | 1.1645 | 2.6978 |

## TEST CASE 2 — VARIABLE L

**Constants:**
R = 10 Ω, f = 2.5 kHz, G = 0.4 × $10^{-6}$ S/km, C = 8 × $10^{-6}$ F/km

| S.No. | L (mH) | $Z_0$ (Ω) | Simulation α (neper/km) | Simulation β (rad/km) | Theoretical α (neper/km) | Theoretical β (rad/km) |
|---|---|---|---|---|---|---|
| 1 | 3.5 | 21.00 − j1.89 | 0.2381 | 2.6392 | 0.2381 | 2.6392 |
| 2 | 6.0 | 27.42 − j1.45 | 0.1823 | 3.4463 | 0.1823 | 3.4463 |
| 3 | 8.5 | 32.62 − j1.22 | 0.1533 | 4.0990 | 0.1533 | 4.0990 |
| 4 | 11.0 | 37.10 − j1.07 | 0.1348 | 4.6617 | 0.1348 | 4.6617 |
| 5 | 13.5 | 41.09 − j0.97 | 0.1217 | 5.1636 | 0.1217 | 5.1636 |

## RESULT:

The impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line is examined at a frequency f=2.5 Khz.

## Test Case 3:

**Explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance values from 8 µF to 16 µF at the interval of 2 µF with R,G and L values constant.**

**AIM**: To explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant.

**PROGRAM:**

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*%pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);

a1=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',a1);

b=imag(P);

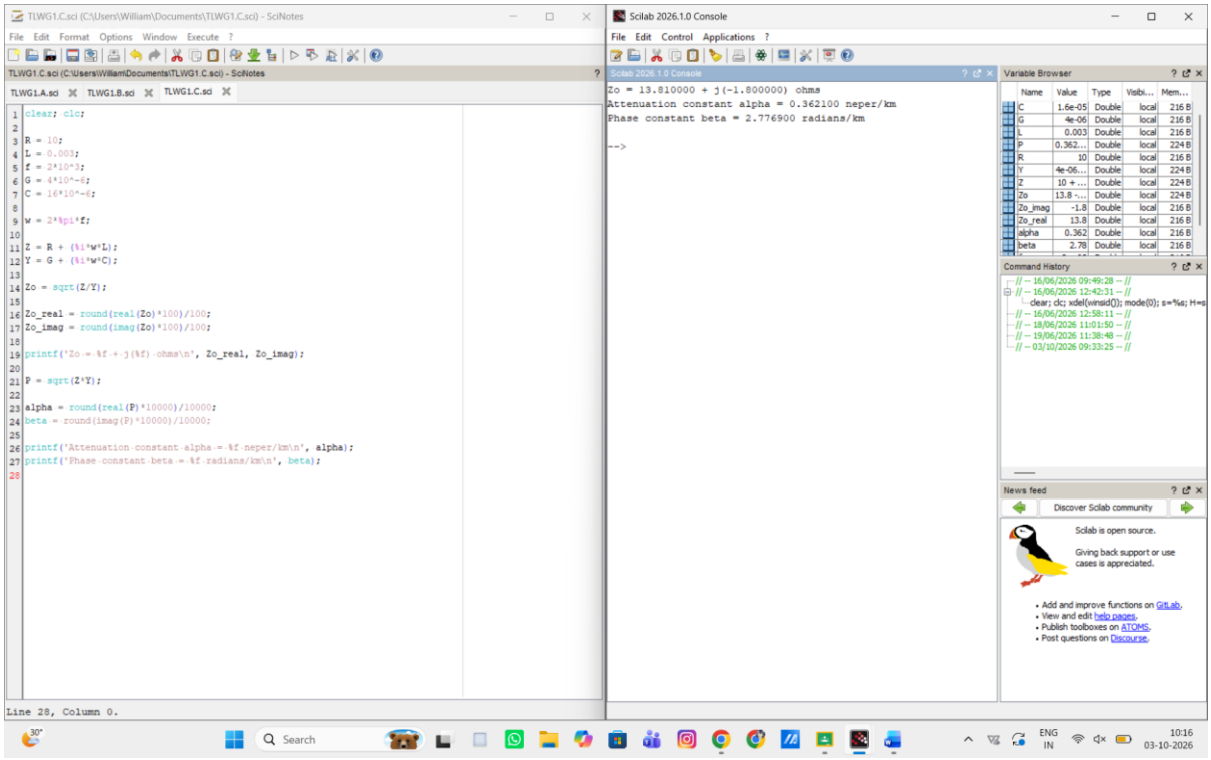
b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);
```

**OUTPUT:**

Variable f

Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).




## TABULATION:

## TEST CASE 3 — VARIABLE CAPACITANCE

Constants: R = 10 Ω, L = 9 mH, f = 2500 Hz, G = 0.4×10⁻⁶ S/km

| S.No. | Input C (μF) | $Z_0$ (Ω) | Simulation α (neper/km) | Simulation β (radians/km) | Theoretical α (neper/km) | Theoretical β (radians/km) |
|---|---|---|---|---|---|---|
| 1 | 8 | 33.562 – j1.185 | 0.1490 | 4.2175 | 0.1490 | 4.2175 |
| 2 | 10 | 30.019 – j1.060 | 0.1666 | 4.7153 | 0.1666 | 4.7153 |
| 3 | 12 | 27.403 – j0.968 | 0.1825 | 5.1654 | 0.1825 | 5.1654 |
| 4 | 14 | 25.370 – j0.896 | 0.1971 | 5.5793 | 0.1971 | 5.5793 |
| 5 | 16 | 23.732 – j0.838 | 0.2107 | 5.9645 | 0.2107 | 5.9645 |

## TEST CASE 3 — VARIABLE FREQUENCY

**Constants:** R = 10 Ω, L = 3 mH, G = $0.4\times10^{-6}$ S/km, C = 8 Mf

| S.No. | Input f (Hz) | $Z_o$ (Ω) | Simulation α (neper/km) | Simulation β (radians/km) | Theoretical α (neper/km) | Theoretical β (radians/km) |
|---|---|---|---|---|---|---|
| 1 | 2000 | 19.532 − j2.546 | 0.2560 | 1.9635 | 0.2560 | 1.9635 |
| 2 | 4000 | 19.407 − j1.281 | 0.2576 | 3.9021 | 0.2576 | 3.9021 |
| 3 | 6000 | 19.384 − j0.855 | 0.2580 | 5.8460 | 0.2580 | 5.8460 |
| 4 | 8000 | 19.376 − j0.642 | 0.2581 | 7.7914 | 0.2581 | 7.7914 |
| 5 | 10000 | 19.372 − j0.513 | 0.2581 | 9.7373 | 0.2581 | 9.7373 |

## RESULT:

The impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line is investigated for various values of frequencies and capacitance values.